Do Younger Directors Make Better Video-Game Movies? Critics' Scores for Films Based on Video Games, 1993–2025

Course example · ECON 3916 Midterm Project · Dr. Richeng Piao

Repository: github.com/RyanPiao/econ3916-midterm-example **Notebook:** colab.research.google.com/github/RyanPiao/econ3916-midterm-example/blob/main/midterm-example-movies.ipynb

Question	Among English-language films based on video games, 1993–2025, are films by younger directors rated better by critics (the Rotten Tomatoes score), holding the release year and the format (live-action or animated) constant?
Hypothesis	Younger directors are more likely to have grown up playing the game they adapt, so they know what its players care about and should make better films: a negative coefficient on age.
Headline finding	A director ten years younger goes with a score 1.7 points higher (95% CI −4.6 to +8.1 points, N = 53). The sign is the one the hypothesis predicts, but the data cannot rule out no association. This is a conditional association, not the effect of having played the game.
Data	One row is one film (53 in the sample, from 68 listed). Scores: Wikipedia's "List of films based on video games", citing Rotten Tomatoes, downloaded 2026-11-03. Directors' birth dates: Wikidata. Budgets in 2024 prices with the CPI-U from FRED.
Method	OLS of the Rotten Tomatoes score (percent of critics' reviews that are positive) on the director's age at release, the release year and an animated indicator, with robust (HC1) standard errors.

1. The Question and the Data

Video-game adaptations have a famously bad record with critics. For years the standard explanation was that the people making them did not understand the games. The directors have changed: many of those hired in the last decade grew up with the games they now adapt. If a director who grew up playing a game makes a better film of it, a studio choosing a director has a reason to look for one. The question matters to the studios making that choice and to the players who wait for these films.

My question is whether, among English-language films based on video games released from 1993 to 2025, films by younger directors are rated better by critics, holding the release year and the format constant. One row is one film. The outcome is the film's Rotten Tomatoes score: the percent of critics' reviews that are positive. The main explanatory variable is the director's age at release, in years. My hypothesis is that younger directors are more likely to have played the game, so I expect a negative coefficient on age. Age is a stand-in for that experience, which I cannot observe directly; Section 5 returns to what that costs.

The data are a product of several measurement decisions. The list of films comes from Wikipedia's "List of films based on video games", which volunteer editors keep up to date and which cites Rotten

Tomatoes for every score. Rotten Tomatoes decides which critics count and whether each review is positive or negative, so a score of 30% means that three in ten counted reviews were positive, not that the film earned 30 out of 100. The score measures critics, not players: a film fans enjoy can still score low. It is also noisier when few critics reviewed the film, and the table reports the number of reviews beside each score. The directors' birth dates come from Wikidata, a volunteer database, matched to the films by the director's name. Budgets, used only in Section 3, come from the same Wikipedia table and are put in 2024 prices with the BLS consumer price index (CPIAUCSL) from FRED.

Building the sample changed the data at each step. The page lists 68 films in two tables, 55 live-action and 13 animated. The title is not a key: Mortal Kombat, Resident Evil and Street Fighter were each remade, so I identify a film by its title and release year. The scores arrive as text ("26% (46 reviews)[4]"), which I split into the score and the number of reviews. Merging the films with the birth dates on the director's name first matched only 58 of 68 films, because films with two directors came out of the web table with the names glued together ("Rocky MortonAnnabel Jankel"); matching on the first director listed fixed that.

Two gaps matter. 4 films have no score at all, because too few critics reviewed them (Far Cry, Dead Trigger, Heavenly Sword, Scarygirl). They are the smallest releases, and a film too obscure to be reviewed is plausibly a weak one, so this gap may depend on the quality the score would have measured: MNAR, which these data cannot test. Its effect is that the sample is the films critics bothered to review. Second, 5 rated films lose the director's age because Wikidata has no birth date. Their scores range widely, and 2 of them are animated, against 15% of the sample, so I treat that gap as MAR given the format, which the model controls for. Films released in 2026 are dropped by a rule written before fitting any model, since their reviews are still coming in. Table 1 shows each step; Table 2 summarizes the 53 films that remain.

Table 1. From 68 listed films to the sample

Step	Films
Films listed	68
Released in 2026 (reviews still coming in)	−6
No Rotten Tomatoes score	−4
No birth date for the director	−5
Films in the sample	53

Steps are applied in order, so a film missing two things is counted once, at the first step it fails.

Table 2. Summary statistics, 53 films, 1993–2025

Variable (units)	Mean	Median	SD	Min	Max	N
Rotten Tomatoes score (% positive reviews)	34.2	29.0	24.5	1.0	100.0	53
Reviews counted	134.3	120.0	83.0	5.0	327.0	53
Director's age at release (years)	45.0	44.0	7.7	30.0	68.0	53

Release year	2011.8	2012	8.9	1993	2025	53
Budget ($m, 2024)	86.5	74.4	59.2	7.5	251.7	50

Budget in millions of 2024 dollars (CPI-U); 50 of the 53 films report one in dollars. Sources: Wikipedia (citing Rotten Tomatoes, Box Office Mojo and The Numbers), Wikidata, FRED. Downloaded 2026-11-03.

2. Descriptive Evidence

Most of these films are reviewed badly. The median score is 29.0%, the mean 34.2%, and only 9 of the 53 films pass 60% (Figure 1). The distribution is skewed right (skewness 0.75), with a long tail of well-reviewed films, which is why Table 2 reports the median beside the mean. A score is already a percentage between 0 and 100, so I model it directly rather than in logs, and check the result with a median regression in Section 4.

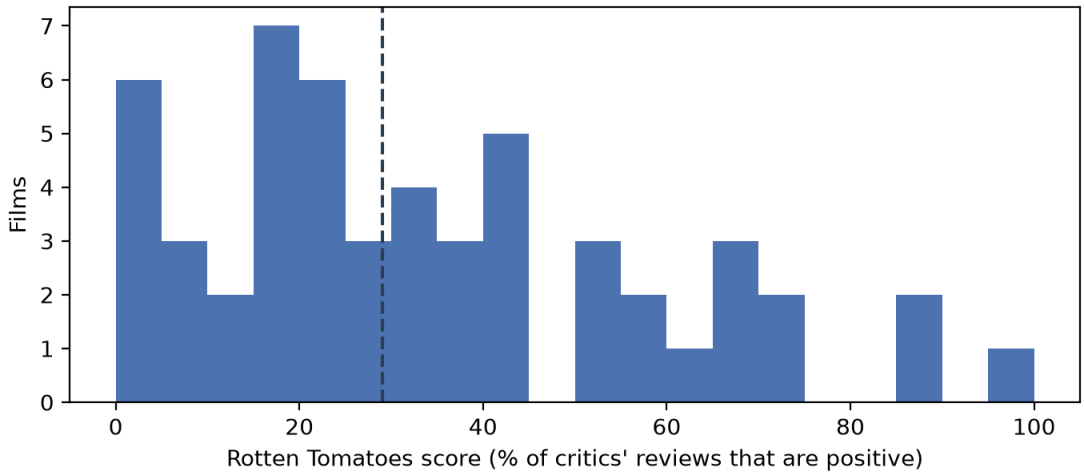


Figure 1. Most video-game adaptations are reviewed badly: the median film (dashed line) has 29.0% positive reviews, and only 9 of the 53 pass 60%. Source: Rotten Tomatoes scores as cited by Wikipedia.

Figure 2 plots each film's score against its director's age at release. Directors range from 30 to 68 years old, and there is no visible slope: well- and badly-reviewed films come from directors of every age, though most of the films come from directors in their forties. The animated films (triangles) sit higher than the live-action ones, which is why the model separates the two formats. Figure 3 shows the pattern that matters most for the model: scores rise with the release year. Films released from 2016 on average 48.3%, against 20.5% for 1993–2005 and 25.5% for 2006–2015. If the directors hired in recent years are also younger, the release year alone could make younger directors look better.

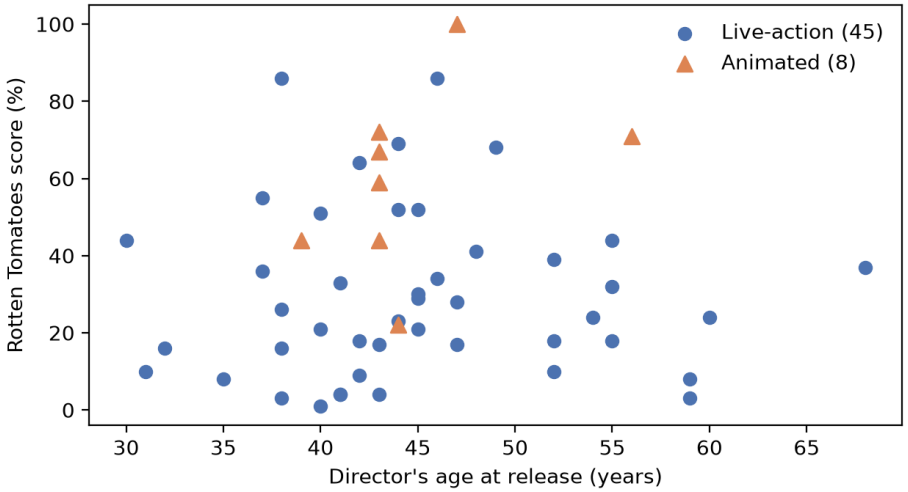


Figure 2. Across directors aged 30 to 68 at release, critics' scores show no visible trend with age, for live-action or animated films (triangles). Source: as Figure 1; ages from Wikidata.

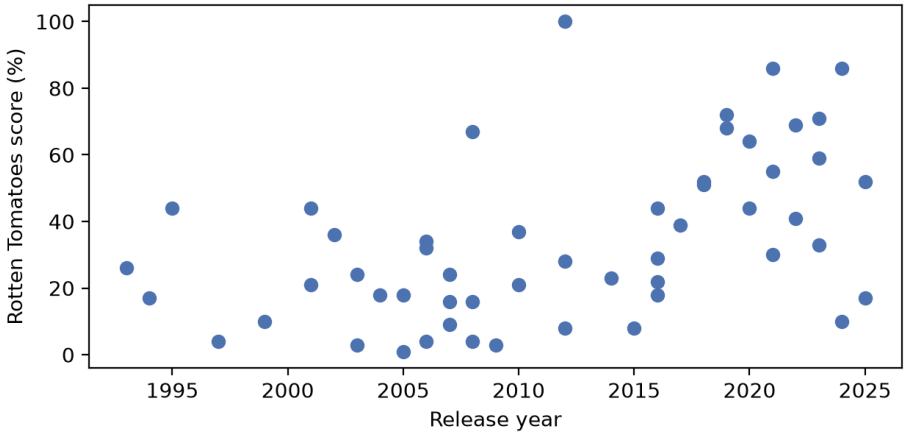


Figure 3. Recent adaptations are rated far better: films from 2016 on average 48.3%, against 20.5% to 25.5% before. Source: as Figure 1.

3. The Model

I fit the model below by OLS, with heteroskedasticity-robust (HC1) standard errors:

$$Score_i = \beta_0 + \beta_1\, Age_i + \beta_2\, Year_i + \beta_3\, Animated_i + u_i$$

The full results are in Table A1 in the appendix; column 2 is the main model. The coefficient on age is −0.17 points per year (standard error 0.32). Holding the release year and the format fixed, a director ten years younger goes with a score 1.7 points higher, with a 95% confidence interval from −4.6 to +8.1 points. The sign is the one my hypothesis predicts, but the interval includes zero: with 53 films, the data cannot tell a small advantage for younger directors from none, or from a small disadvantage. Without any controls (column 1) the coefficient is +0.05 points per year, the wrong sign for the hypothesis; holding the release year fixed is what turns it negative.

The second coefficient I interpret is the release year: +1.21 points per year (95% CI +0.56 to +1.85). Holding the director's age and the format fixed, a film released ten years later scores about 12 points higher, and this interval excludes zero. The animated indicator is +26.1 points (95% CI +8.1 to +44.0), though only 8 films are animated. The R^2 is 0.38.

Figure 4 draws the assumptions behind the model, and each control has a role in it:

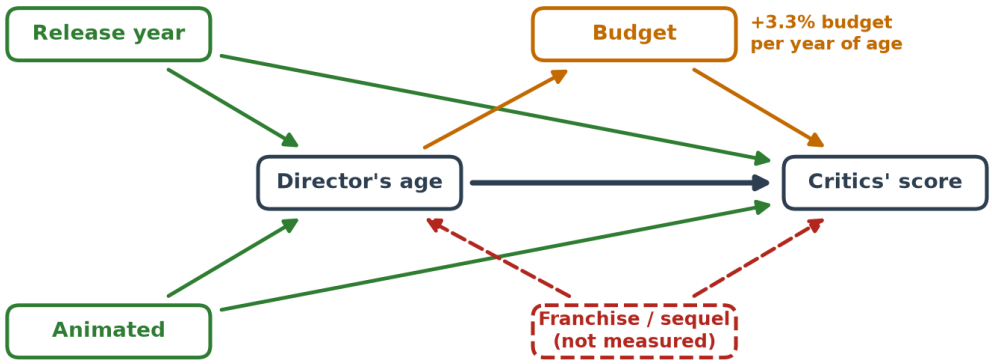


Figure 4. The causal graph I assume. Release year and format point into both the director's age and the score, so the model controls for them. Budget sits between age and the score, so it stays out. Franchise and sequel status, which I cannot measure, leaves a path open (Section 5).

- **Release year** blocks a confounding path. Critics' scores for these films rose sharply over three decades (Figure 3), and studios may have hired younger directors in the same years; the column 1 to column 2 change shows the path is real. It would be a bad control only if a director's age could change a film's release year, which it cannot.
- **Animated** also blocks a confounding path: animated adaptations are made by different directors and reviewed on a different curve. The studio picks the format before hiring the director, so age cannot cause it.
- **Budget** is left out on purpose. Older directors get bigger budgets (3.3% per year of age, holding year and format fixed), and the money can change how good a film is, so budget is a mediator of age's total effect and a bad control for my question. Columns 3 and 4 compare the same 50 films without and with it: the age coefficient moves from −0.27 to −0.47 points per year (95% CI −1.08 to +0.14), and neither is distinguishable from zero. Column 4 answers a different question, the direct path only; my headline is column 2. The number of reviews is not a control either, because it depends on how much attention the film drew.

4. Diagnostics

I checked the main model with the tests from Topic 14. The residual plots in Figure A1 show no curve against the fitted values or against age, so a straight line in age is a reasonable shape. The spread of residuals is wide, a standard deviation of 19.8 points, which matches an R^2 of 0.38: critics' scores vary a great deal around the model.

Constant variance: the Breusch–Pagan test gives $p = 0.14$, no evidence of unequal variance. With 53 films the test has little power, so I lean on the plots and report robust standard errors anyway. The

choice is visible in the age coefficient: its standard error is 0.36 with classical errors (95% CI −0.90 to +0.56) and 0.32 with HC1 (−0.81 to +0.46). Both include zero, so the conclusion does not depend on it.

Collinearity: the variance inflation factors are 1.03 for age, 1.05 for the release year and 1.02 for the animated indicator. In these films the director's age is almost unrelated to the year and the format, so the wide interval on age comes from the small sample and the noisy scores, not from overlap among the regressors.

Influence: using Cook's distance with the cutoff 4/n = 0.075, 5 films stand out. The three largest are Resident Evil: Damnation (0.208), Ratchet & Clank (0.163) and Mortal Kombat (0.121). Resident Evil: Damnation's score of 100% rests on 5 reviews. A score from so few reviews is mostly noise, so I report the estimate without the 3 films reviewed fewer than 20 times (Resident Evil: Degeneration, Resident Evil: Damnation, Resident Evil: Death Island): the age coefficient goes from −0.17 to −0.27 points per year (standard error 0.34, N = 50).

The median: because the score is skewed, I also fit the same model as a median regression. The age coefficient there is +0.57 points per year (95% CI −0.47 to +1.61): the point estimate changes sign, and the interval still includes zero. Every version of the model gives the same reading, no clear association between the director's age and the critics' score.

5. Threats and Conclusion

For the age coefficient to be causal, nothing else that moves a film's score could vary with its director's age once the year and the format are fixed, and franchises break that. Resident Evil, Mortal Kombat and Sonic each have several films here, and studios hand the next film in a series to a director they already trust. Whether that pushes the coefficient up or down depends on who they trust, and my data have no franchise or sequel variable to block the path.

The bigger threat is measurement. My hypothesis is about whether the director played the game; age is only a stand-in for it. A director of 50 may have played Mortal Kombat in 1992, and a director of 35 may never have touched the game they adapt. A sharper measure would be the director's age when the game first came out, but the source-game release years I could get from Wikidata cover only about half of these films and date a series by whichever entry a film cites. The scores also measure critics, not the players the hypothesis is about.

So the estimate is a conditional association, not the effect of having grown up with the game. Among these 53 films, holding the release year and the format fixed, a director ten years younger goes with a score 1.7 points higher, with an interval from −4.6 to +8.1 points. A reader should take from it that the data do not support the hypothesis strongly in either direction, while the release year matters far more than the director's age. With another month I would date each source game by hand, use the director's age when it came out, add a franchise flag, and add audience scores to see whether players and critics agree.

References

Federal Reserve Bank of St. Louis. Consumer Price Index for All Urban Consumers: All Items in U.S. City Average (CPIAUCSL), from the U.S. Bureau of Labor Statistics. FRED. Retrieved 2026-11-03.

Wikidata. Date of birth (P569) for each director, retrieved through the Wikidata API on 2026-11-03.

Wikipedia contributors. “List of films based on video games.” Wikipedia. Retrieved 2026-11-03. Scores as cited there from Rotten Tomatoes; budgets from Box Office Mojo and The Numbers.

Appendix

Table A1. Rotten Tomatoes score regressed on the director's age and controls

	(1) Age only	(2) Main	(3) Main, budget films	(4) Adds budget
Director's age (years)	0.05	-0.17	-0.27	-0.47
	(0.37)	(0.32)	(0.34)	(0.31)
Release year		1.21	1.27	1.34
		(0.33)	(0.34)	(0.33)
Animated (0/1)		26.09	13.21	10.72
		(9.16)	(8.41)	(7.39)
log budget				6.20
				(5.21)
Constant	31.94	-2387.72	-2517.76	-2664.78
	(17.27)	(660.85)	(677.86)	(666.15)
N	53	53	50	50
R^2	0.00	0.38	0.32	0.36

Note: Dependent variable: Rotten Tomatoes score, percent of critics' reviews that are positive. OLS with robust (HC1) standard errors in parentheses. Column 2 is the main model. Column 3 re-estimates it on the 50 films with a dollar budget, so that column 4, which adds the log budget (a mediator, Section 3), compares the same films.

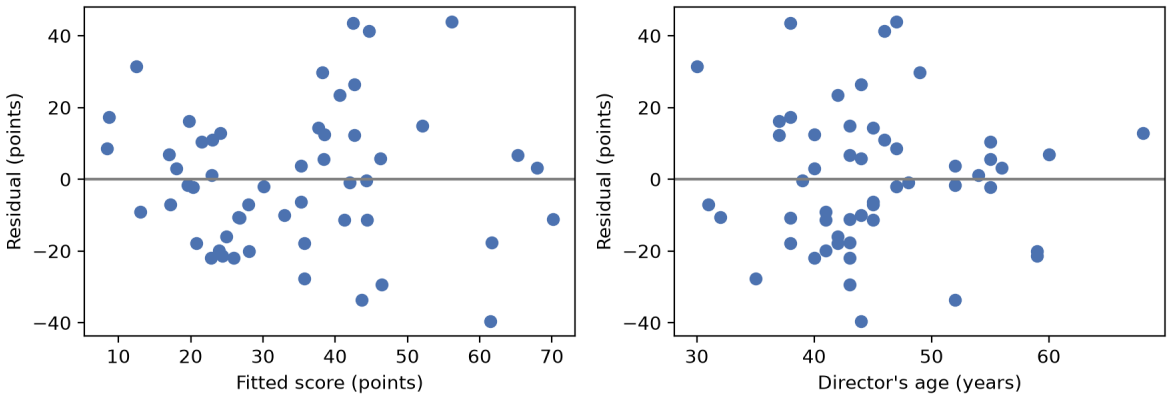


Figure A1. The main model's residuals show no curve against the fitted values (left) or the director's age (right); the spread is wide throughout.

Reproducing the numbers. Every number in this report is printed by the last code cell of the notebook in the repository, "Numbers my report quotes". The README gives the download step and the order to run the cells.